

20 MIN SOLO WORK · 5 MIN PRESENT · 5 MIN Q&A

Walk the Product Engineer's Loop

One small feature. Six steps. 20 minutes to work it out, 5 minutes to defend it live.

1 Understand the problem

2 Know the user

3 Define the outcome

4 Build the smallest thing

5 Measure real usage

6 Learn and iterate

Today's feature: “Warn me before my account balance runs low.” You have 20 minutes, alone, to take this one raw ask through all six steps of the loop on the worksheet slides that follow. Then you present your answer for 5 minutes and take 5 minutes of questions from the room.

BEFORE YOU START

How this exercise works

Same six steps as the loop you just saw — now applied on paper, fast, to a single small feature

20-MINUTE TIMEBOX, THEN 5 + 5

0:00–0:02	Read the brief below. Write nothing yet — just understand what you're being asked to build.
0:02–0:05	Step 1 · Understand the problem
0:05–0:08	Step 2 · Know the user
0:08–0:11	Step 3 · Define the outcome
0:11–0:14	Step 4 · Build the smallest thing
0:14–0:17	Step 5 · Measure real usage
0:17–0:20	Step 6 · Learn and iterate
0:20	Timer stops — worksheet closed, no more writing
+5 min	Present your six answers as one story (cue card ahead)
+5 min	Take questions from the room (guide ahead)

THE RAW ASK YOU'RE GIVEN

“Customers keep getting hit with overdraft fees because they don't realise their balance is low. Add a low-balance alert to the app.” — from a product manager's email

System: the Personal Banking Management System you've already seen — accounts, transactions, notifications.

You already know this app: it has accounts, balances, transaction history and a notification service. Don't redesign the whole app — apply the loop to this one ask.

RULES OF THE EXERCISE

- Work alone. Write words and short phrases only — you'll explain your reasoning out loud later, not on the page.
- Do the steps in order. Step 4 will tempt you to jump to a screen — resist it until step 3 is done.
- 3 minutes per step is tight on purpose. A fast, honest answer beats a polished, late one.
- There is no single correct feature design — but there is a correct understanding of what each step is for, and that's what your 5 minutes of questions will test.



Why individually first Everyone skips steps under time pressure — usually straight to step 4. Doing this alone, then standing behind it in Q&A, shows you exactly which step YOU skip.

Understand the problem

A feature request is not a problem statement. Before anything else, separate the symptom, the underlying problem, and the proposed solution someone already jumped to.

YOUR TASK — WORDS AND PHRASES ONLY, ~3 MIN TOTAL

1

1

Symptom (what's happening) — Write the one sentence from the brief that describes the pain, in your own words.

2

Problem (why it happens) — Why does this actually happen? (Hint: it's not 'there's no alert' — that's the solution. What's the real cause?)

3

Proposed solution (what was asked for) — Name the specific thing the PM asked for. Then write one alternative solution to the SAME problem that isn't an alert.

4

Worth solving? — In one sentence, say why this problem is worth the team's time right now.



FUNDAMENTALS CHECK

Which one of these is a genuine **PROBLEM** statement, not a solution in disguise?

A. "We need a low-balance alert."

B. "Customers don't know their balance is low until it's too late to act."

C. "Add a red banner when balance < ₹500."

D. "Build a notifications microservice."

Know the user

‘The user’ is not one person. Name who is actually affected, what they know, what device and moment they're in, and what happens if you get the actor wrong.

YOUR TASK — WORDS AND PHRASES ONLY, ~3 MIN TOTAL

1

Primary actor — Who hits this problem? Give them a name, an age range and one relevant habit (not a full persona bio).

2

The moment — Where are they, and what are they doing, in the seconds before this problem bites them?

3

What they already know — What do they understand about their own banking today? (e.g. do they check balance often, or rarely?)

4

A second actor — Name one OTHER actor this feature touches who is NOT the customer (a system, a support agent, a bank team).

?

FUNDAMENTALS CHECK
A teammate says: “Our user is anyone with a bank account.” What's wrong with that, and what should they say instead?

Define the outcome

An outcome is a measurable change in the world, not a shipped feature. Good outcomes have a leading signal you can see fast and a lagging signal that proves it mattered.

YOUR TASK — WORDS AND PHRASES ONLY, ~3 MIN TOTAL

3

1 **The outcome (not the output)** — Complete: 'If this works, fewer customers will ____.'

2 **Leading metric** — What could you measure within DAYS of launch that suggests it's working?

3 **Lagging metric** — What would you measure a QUARTER later that proves it actually reduced harm (e.g. overdraft fees)?

4 **Guardrail** — What must NOT get worse while you chase this outcome? (e.g. alert fatigue, opt-outs)

? FUNDAMENTALS CHECK

Which of these is a **LEADING** metric for this feature, not a lagging one?

A. Overdraft fees charged per customer per quarter

C. Customer churn after 6 months

B. % of customers who set a balance threshold in week 1

D. Net Promoter Score at year end

Build the smallest thing

The smallest thing is the thinnest slice that a real customer could use end-to-end today — not a prototype, not a cut-down UI, a real (if narrow) path through the system.

YOUR TASK — WORDS AND PHRASES ONLY, ~3 MIN TOTAL

4

1 **One-sentence MVP** — Describe the smallest version of this feature a customer could actually use tomorrow.

2 **The main flow (4–6 steps)** — List the steps from trigger to outcome — nothing more than the happy path.

3 **Deliberately deferred** — Name TWO things a fuller version would have, that you are leaving out on purpose. Why is each safe to defer?

4 **The one thing that must not break** — What is the single failure that would make this feature actively harmful rather than just incomplete?



FUNDAMENTALS CHECK

Four teams propose a 'smallest thing' for this feature. Which is actually right-sized?

- A. A settings page with 6 alert types, SMS/email/push choice, and a snooze schedule
- C. A slide deck describing how the alert would work

- B. A fixed ₹500 threshold alert, push notification only, on one account type
- D. A backend rule engine with no way for the customer to see or change anything

Measure real usage

Instrumentation without a threshold is just noise. Decide, before launch, what number you'll watch and what value would count as 'this is working' versus 'this failed'.

YOUR TASK — WORDS AND PHRASES ONLY, ~3 MIN TOTAL

5

1

The one event to log — What single user action, logged with a timestamp, tells you this feature is being used?

2

The 'it's working' threshold — Referring back to your Step 3 leading metric — what number, by when, means success?

3

The 'it's failing' signal — What number or pattern would tell you to stop and rethink, not just wait longer?

4

A vanity trap — Name one number about this feature that would look impressive but tell you nothing useful. Why is it useless?



FUNDAMENTALS CHECK

"40,000 alerts were sent last month" — is this a useful measure of success on its own?

Learn and iterate

Real data rarely says ‘ship it’ or ‘kill it’ cleanly. Learning means turning a number into a specific next hypothesis — not a shrug, and not blind persistence.

YOUR TASK — WORDS AND PHRASES ONLY, ~3 MIN TOTAL

6

- 1 The result you're given** — After 3 weeks: 62% of customers set a threshold; of those, only 18% opened the alert when it fired. -----
- 2 What this tells you** — In one sentence, what does the gap between 62% and 18% suggest is actually broken? -----
- 3 Next hypothesis** — Write your next ‘We believe ____ will ____; we'll know when ____’ for the next iteration. -----
- 4 What you would NOT do next** — Given this data, name one tempting next step that would be the WRONG move, and say why. -----



FUNDAMENTALS CHECK

Given the 62%/18% result above, which is the strongest next move?

- A. Add five more alert types so customers have more to choose from
- B. Change the alert channel/wording and test if open rate improves before building anything else
- C. Declare the feature a failure and remove it
- D. Wait another quarter for more data with no changes

WHEN THE TIMER STOPS • PRESENT • 5 MINUTES

Turn your six answers into a 5-minute story

Walk the room through the loop in order — talk from memory, don't just read the worksheet aloud

1

Problem

0:00–0:30

State the symptom, then the real cause, in one sentence each.

2

User

0:30–1:00

Name the actor and the exact moment this hits them.

3

Outcome

1:00–1:45

Your leading metric, your lagging metric, and the guardrail.

4

Smallest thing

1:45–3:15

The main flow, then what you deliberately cut and why that's safe.

5

Measurement

3:15–4:00

The one event you'd log, and your working / failing thresholds.

6

Learn & iterate

4:00–5:00

Given the sample data, your next hypothesis — and what you'd NOT do.



Presenting is not reading Explain WHY you answered that way — especially step 4 (what you cut and why) and step 6 (why that next hypothesis, not another). That reasoning is what the room will question.

THEN · AUDIENCE Q&A · 5 MINUTES

Ask the question that tests the step, not the idea

Everyone in the room asks — pick a step, not the feature concept, and probe it

1

Problem

"What would you actually observe if this problem stopped existing?"

2

User

"Would every customer feel this the same way — who wouldn't?"

3

Outcome

"Could your leading metric move by Friday, or is it secretly lagging?"

4

Smallest thing

"What did you leave out, and why is that specific thing safe to cut?"

5

Measurement

"What number would make you stop, not just wait longer?"

6

Learn & iterate

"Does your next hypothesis fix the exact gap in the data, or just add scope?"



For the presenter If a question stumps you, don't bluff. Say exactly what you'd need to find out and how — that's still a correct fundamentals answer.



For the audience One good, specific question beats three vague ones. Aim your question at ONE row from the peer-feedback rubric (next slide).

DURING Q&A

Peer feedback — rate what you just heard

Everyone fills this in for the presenter while asking and listening — hand it to them after their round

Step	They sounded like they understood this step if...	The red flag to listen for	Your rating
1	Your problem sentence has no feature name in it anywhere.	Writing the solution again and calling it the problem.	✓ ? ✕
2	You can say WHO and WHY, not just ‘the user’.	Treating every customer as identical.	✓ ? ✕
3	You named a metric that moves in days, and one that proves value later.	Confusing ‘we shipped it’ with ‘it worked’.	✓ ? ✕
4	Your slice is smaller than what you first imagined — and still usable.	Building the settings page before the core flow.	✓ ? ✕
5	You picked a threshold BEFORE seeing any data.	Deciding what ‘good’ looks like after seeing the number.	✓ ? ✕
6	Your next hypothesis targets the specific gap the data showed.	Adding more features instead of fixing the one that's broken.	✓ ? ✕



After each round Hand the presenter one filled-in row — your rating plus a specific reason. That written note is more useful to them than anything said out loud in 5 minutes.



Remember The six steps are a loop, not a checklist you complete once. Step 6 feeds a new Step 1 — the next iteration starts from what you just presented.